

Orlando Javier Marrero González

AI Engineer | Software Engineer

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SUMMARY

Junior software engineer with 2 years of experience building desktop and web applications, with hands-on experience integrating small, on-device language models into a production speech-transcription desktop system as part of an undergraduate thesis developed at Ingenius. Comfortable across the full stack — from desktop app UI and internationalization to backend services in Java/Spring Boot — with a growing specialization in local LLM inference, NLP pipelines, and prompt engineering. Currently deepening agentic AI skills (LangGraph) to move further into AI engineering roles.

SKILLS

AI & NLP: On-device/local LLM inference, small language models (Phi-4-mini), model quantization (GGUF, Q4_K_M), llama.cpp / llama-cpp-python, prompt engineering, Map-Reduce summarization pipelines, post-ASR text correction, hallucination-risk mitigation, PyTorch, statistical validation. Currently learning: LangGraph, multi-agent orchestration.

Backend: Python, Java, C, C++, Spring Boot, PostgreSQL, SQLite, REST APIs, Nest.js, FastAPI

Frontend & Desktop: Vue.js, JavaScript, Flet (Python desktop UI framework), UI development, internationalization / i18n (Spanish, English, French)

Tools: Git, Agile teamwork, Docker

EXPERIENCE

Ingenius — Ingeniero de Software Junior

2024 – 2026 (~2 years) | La Habana, Cuba

- Developed and maintained the **UI of Notus**, the company's desktop speech-to-text transcription product built on Whisper and WhisperX, using the Flet (Python/Flutter) framework.
- Implemented **internationalization (i18n)** support for the app in Spanish, English, and French, enabling multi-language use for end users.
- Undergraduate thesis project: designed and integrated **two on-device NLP modules** into Notus, both running the small language model **Phi-4-mini fully locally** (Q4_K_M quantization, GGUF format, via llama.cpp, ~3GB VRAM footprint).
- Built a **post-ASR linguistic correction pipeline** (punctuation, casing, homophone correction) using segmentation, inference, and post-generation validation to control model hallucination risk.
- Extended the existing summarization module with **3 styles × 3 length levels** using a **Map-Reduce** pipeline, and integrated both modules into Notus's existing GUI and architecture.
- Validated both modules with non-parametric statistical tests (Wilcoxon signed-rank) on a custom 10-audio Spanish corpus: the correction module **significantly restored punctuation** with minimal impact on word/character error rate, and the summarization module reached **high semantic similarity** against a larger reference LLM across all styles and lengths.

Technologies: Python, PyTorch, Flet, Phi-4-mini, llama.cpp, Whisper/WhisperX (integration), i18n.

ETECSA — Full-Stack Developer (contract)

Aug 2024 – Nov 2024 | La Habana, Cuba

- Collaborated on the **database design and backend** of a digital-transformation assessment platform that scores companies across dimensions, perspectives, and areas from structured questionnaires, as part of a 3-developer team.
- Built the **desktop version** of the platform's user interface using **Java Swing (Java 22)**.
- Built the **web version** of the platform's user-facing interface using JavaScript/Vue.js, in addition to backend development.

Technologies: Java, Java 22, Java Swing, Spring Boot, PostgreSQL, SQLite, REST APIs, JavaScript/Vue.js.

EDUCATION

B.Sc. Software Engineering — Ciudad Universitaria José Antonio Echeverría (CUJAE) | Feb 2023 – Jun 2026 | La Habana, Cuba

LANGUAGES

Spanish — native. English — A2 (certificate issued by the university).

SELECTED PROJECTS

On-Device NLP Modules for Notus — Undergraduate Thesis

2026 | *Ingenius*

- Designed and implemented a **post-ASR linguistic correction module** for Notus, fixing punctuation, capitalization, and homophone errors in Whisper/WhisperX transcriptions using the local SLM Phi-4-mini.
- Built the correction pipeline around **text segmentation, local inference at low temperature (0.1), and a post-generation validation/filtering step** to reduce hallucination risk.
- Extended Notus's automatic summarization module to support **3 styles × 3 length levels**, using a **Map-Reduce** pipeline to handle Phi-4-mini's limited context window.
- Applied the **Facade design pattern** over the processing package to cleanly integrate both modules into Notus's existing Flet-based GUI and architecture.
- Ran a rigorous statistical evaluation on a custom 10-audio Spanish corpus: used the **Wilcoxon signed-rank test** to compare configurations and validate improvement over uncorrected Whisper output, and **Kruskal-Wallis + Mann-Whitney U** (with correction for multiple comparisons) plus **BERTScore F1** to benchmark Phi-4-mini's summary quality against larger reference LLMs.
- Results: the correction module **significantly restored punctuation** ($p < 0.001$) with minimal added word/character error rate versus raw Whisper output, and the summarization module reached **high semantic similarity** versus a larger reference LLM across all styles and lengths, while running fully on-device.

Technologies: Python, PyTorch, Flet, Phi-4-mini, llama.cpp / llama-cpp-python, Whisper, WhisperX.

LangGraph Agent — Coming Soon

- In progress: an agentic project built with LangGraph, focused on multi-agent orchestration and tool use. Details and repository link will be added once complete.